DECA Section 1

2025年11月11日 星期二 08:27

Digital Electronics & Computer Architecture Lab. S1.
24 October, 2025.
Lab, Charles Wu.

Reference: DECA Section 1.

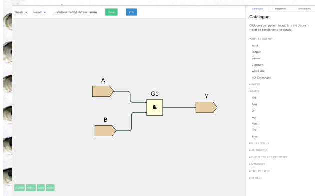
Aim. Learn how to use ISSIE, design multiplexers & simple ALU.

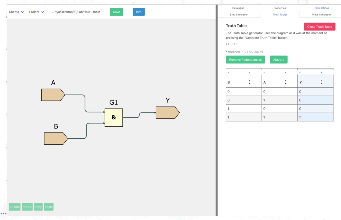
Procedure,

- Building a simple AND Gate and simulate it works by truth table:

what expect to get: $Y = A \times B$.

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1





Which is true

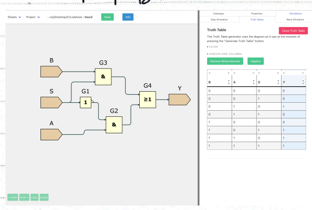
- Building a 2-input multiplexer using AND; OR; and inverter.

A multiplexer controls which input to use when there are several inputs and sends the right one to output.

It has boolean expression: $Y = A \cdot \bar{S} + B \cdot S$ (2-input)

Expect:

S	Y
0	A
1	B



- Trying hierarchical design and create a 4-input multiplexer.

HIDE/UN-HIDE COLUMNS

Back to Numeric Table | Algebra | Key

Algebraic Inputs: D C B A

A	B	C	D	S0	S1	Y
A	B	C	D	0	0	A
A	B	C	D	0	1	C
A	B	C	D	1	0	B
A	B	C	D	1	1	D

Equation: $Y = A\bar{S_0}\bar{S_1} + BS_0\bar{S_1} + C\bar{S_0}S_1 + DS_0S_1$.

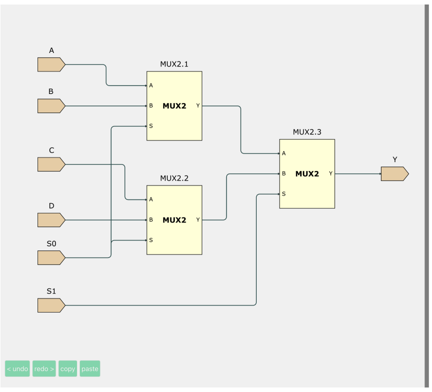
~~Except~~ Expect.

S_0	S_1	Y
0	0	A
1	0	B
0	1	C
1	1	D

- This can be done by using 2-input (MUX2)s as sub-sheet.

A and B can go through MUX2; C and D go through another MUX2;

then the results of AB and CD goes through a third MUX2 to finally locate an accurate one.



the "Generate Truth Table" button.

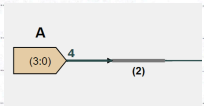
▶ FILTER

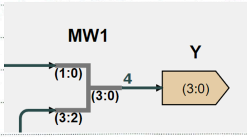
▶ HIDE/UN-HIDE COLUMNS

A	B	S0	C	D	S1	Y
0	0	0	0	0	0	0
0	0	0	0	0	1	0
0	0	0	1	0	0	0
0	0	0	1	0	1	1
0	0	0	0	1	0	0
0	0	0	0	1	1	0
0	0	0	1	1	0	0
0	0	0	1	1	1	1
0	1	0	0	0	0	0
0	1	0	0	0	1	0
0	1	0	1	0	0	0
0	1	0	1	0	1	1
0	1	0	0	1	0	0
0	1	0	0	1	1	0
0	1	0	1	1	0	0
0	1	0	1	1	1	1

For SELECT, one select controls up to $2^1 = 2$ inputs
2 $2^2 = 4$
3 $2^3 = 8$
...

- Learning to use 'MergeWires' & 'BusSelect' for entering 4-bit numbers.





One technique is that we can only test AB in the wiring to check for MUX2×4, as MUX2 functions well.

Expect:

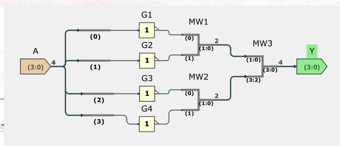
A	B	S	Y
0001	1110	0	0001
0010	1101	1	1101
1000	0111	0	1000
0100	1011	1	1011

, which is correct.

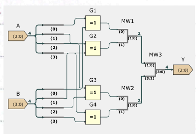
...

- Challenge: Building a simple ALU for 2 inputs of 4-bits, including: NOT A ; A OR B ; A AND B ; A XOR B.
Just merge what we've learned, build 4 operations in sub-sheet first, then use them in the 'main', finally, use a MUX4 to merge them and select needed data.

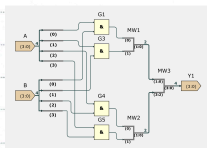
Inverter:



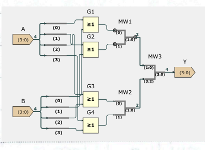
2×4 XOR.

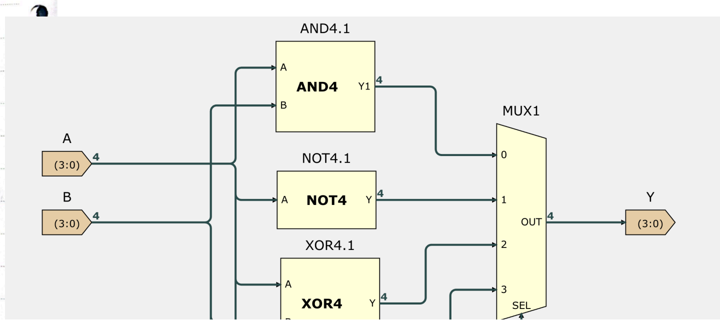


2×4 AND.

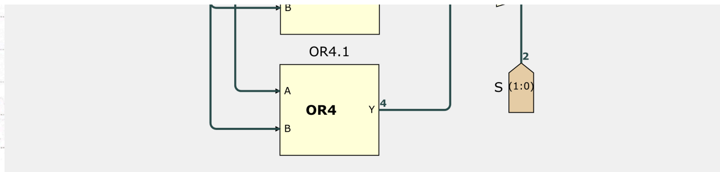


2×4 OR.





After merging :



Do a random test, I use.

A	B	S_0	S_1	Y
1010	0110	0	0	0101
1010	0110	0	1	1110
1010	0110	1	0	0010
1010	0110	1	1	1100

which gives correct results of NOT A, A|B, A&B, A⊕B in order.

[illegible].